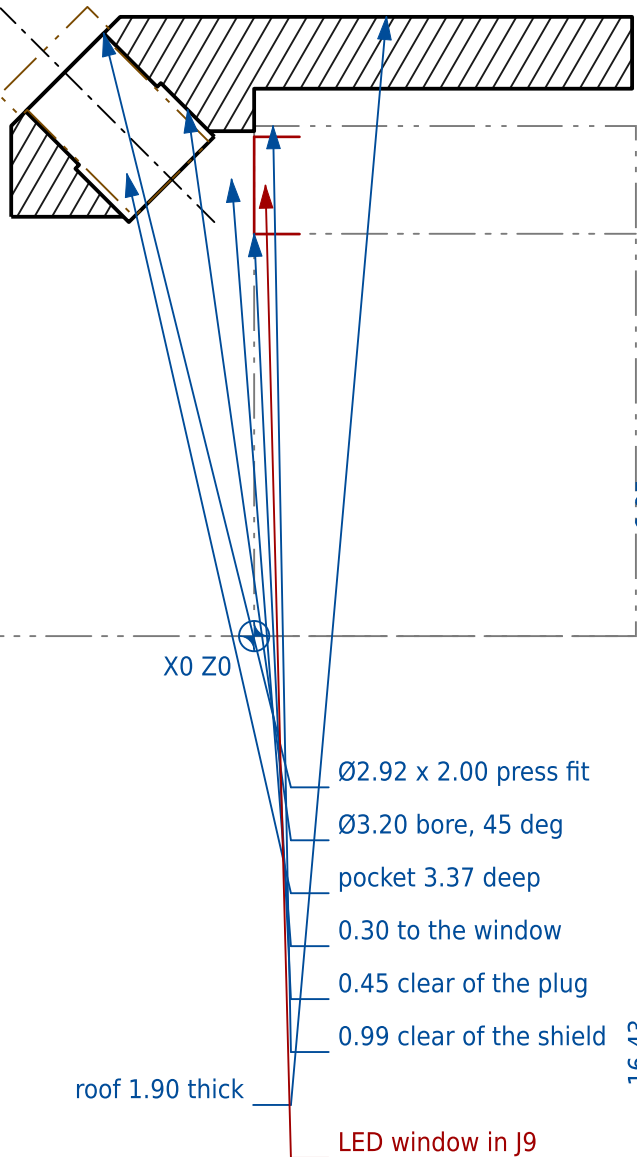
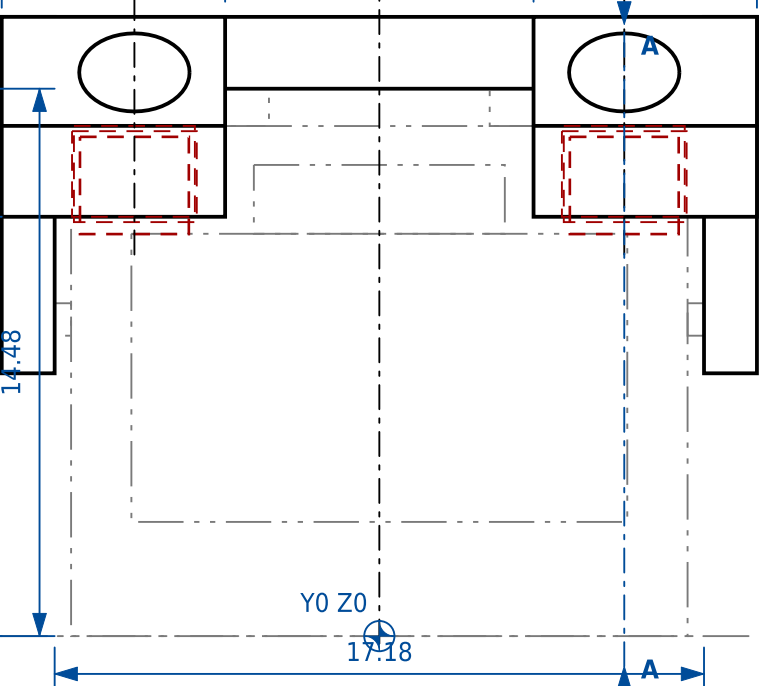


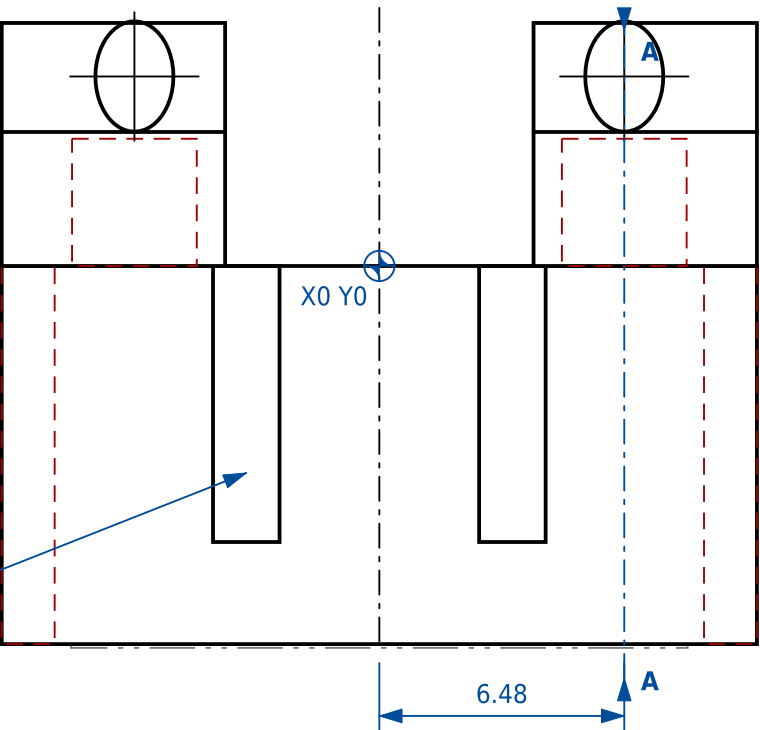
SECTION A-A



FRONT ELEVATION



PLAN



BORE SCHEDULE, DERIVED

ID	Y mm	TIP X / Z mm	DIA mm	PRESS mm	PRESS LEN	DEG
B1, B2	+/-6.48	-2.18 / 12.08	3.20	2.92	2.00	45

THE JACK, J9 08B0-1X1T-36-F

FEATURE	Y mm	Z mm	READ AS
LED window, 0.64 proud	5.04..7.92	10.63..13.20	MEASURED
Plug aperture	+/-6.56	3.02..10.64	MEASURED
Latch keyway, widest	+/-3.32	10.64..12.46	MEASURED
Shield	+/-8.15	0..13.49	STATED
EMI spring, side	+/-9.18	7.95..8.80	MEASURED
EMI spring, top	+/-2.92..4.12	to 14.52	MEASURED

LEGEND

- The printed part
- Hidden detail: bores and the pocket
- The RJ45 J9 and its EMI springs, adjacent parts
- The two Bivar light pipes, bought

NOTES (continued)

- The press fit is Bivar's $\text{Ø}2.92 +0.08/-0.05$ and that governs, not the general tolerance; ream it if the printed hole comes out undersize.
- A plug and a plain boot pass under the cheeks; a snagless boot standing proud of the plug's own top face within 7 mm of the jack will foul them.
- STATED is Bel's own dimension, ± 0.254 ; MEASURED is read off its geometry, ± 0.20 ; DERIVED is worked out from those and Bivar's pipe.

SOURCES

- Board schematic: https://digilent.com/reference/_media/reference/programmable-logic/arty-a7/arty_a7_sch.pdf - Digilent, Arty A7 rev E.0, sheet 8 of 12 (ETHERNET): J9 is 08B0-1X1T-36-F.
- Jack drawing: <https://www.belfuse.com/media/drawings/products/magjack%20ICMs/dr-mag-08b0-1x1t-36-f.pdf> - Bel 08B01X1T36-F rev E; MEASURED figures are scaled off its front and side views.
- Board drawing: https://digilent.com/reference/_media/reference/programmable-logic/arty-a7/arty_a7.zip - Digilent, Arty_A7_DXF.DXF: board locks 16.104 apart against Bel's 16.13; centre plane $y = 44.0$, and through Bel's 7.75 the front face $x = -0.101 \pm 0.628$, which is how far Digilent's own plot of the jack disagrees.
- Light pipe drawing: https://www.bivar.com/parts_content/Datasheets/PLP2-XXX.pdf - Bivar PLP2-XXX rev Y: $\text{Ø}2.8$ pipe on $\text{Ø}3.1$ ribs, $\text{Ø}3.3$ flange, $\text{Ø}0.115$ in hole, 0.047-0.093 in panel.

NOTES

- First angle; SECTION A-A is on the right-hand bore's axis. Origin on the jack's centre plane, in its front face, at the board's top surface: X back, Y across, Z up. Symmetric about $Y = 0$.
- Two Bivar PLP2-4MM pipes in clear polycarbonate, pressed in from the facet until the flange seats. Each lens looks up and forward at 45 degrees, so a camera looking straight down sees it obliquely.

NOTES (continued)

- J9's LED windows stand 0.64 mm proud of the face the cheeks land on, so the pipe tips are set from the window, at maximum material.
- Nothing holds the adapter on but J9's own side EMI springs, 1.4 ± 0.51 proud of a 16.31 ± 0.254 shield, bearing on the skirts: deflected 0.33 mm at minimum material and 1.6 at maximum, the shield never touching.
- Print upside down, on the top face: the facets and the bores are then 45 degree overhangs and nothing needs support.

ARTY A7 ETHERNET LIGHT PIPE

Clips over J9 and carries its two LEDs to a face you can see from above

DRAWN generated	VERSION v0.0-304-gfe485e3	UNITS mm	SCALE 5:1
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-LP-ARTY	REV A
MATERIAL 3D printed, opaque	GENERAL TOLERANCE printed ± 0.20 bores: see note on the press fit		PROJECTION